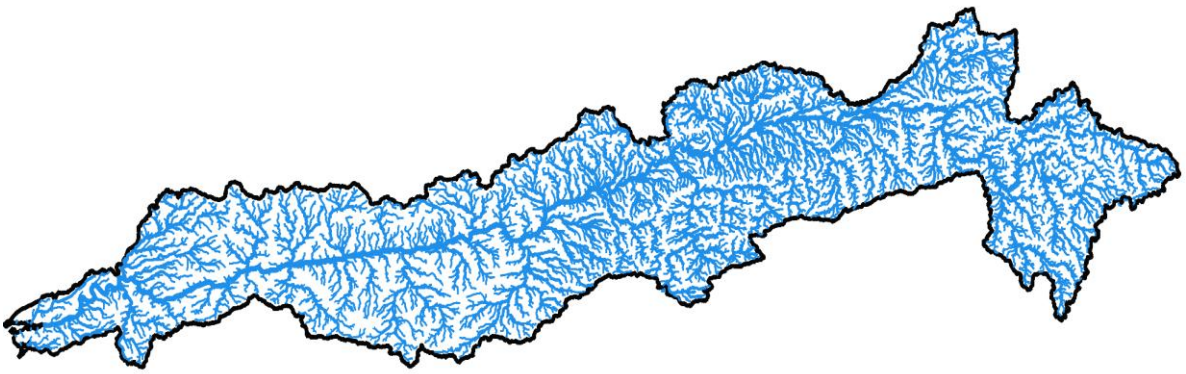
National River Conservation Directorate
Ministry of Jal Shakti,
Department of Water
Resources,
River Development & Ganga Rejuvenation
Government of India

Identification of Major Threats to Biodiversity, Management, and Conservation



June 2026

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National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and Ministry of Jal Shakti providing financial assistance to the State Government for conservation of Rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of Rivers in identified polluted River stretches based on proposals received from the State Governments/ local bodies.

www.nrcd.nic.in

Centers for Narmada River Basin Management and Studies (cNarmada)

The Center for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and the River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the center serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement.

www.cnarmada.org

Centers for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in River and water science. The Center is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as think-tank in implementation and dynamic evolution of Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India.

www.cganga.org

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This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCD) in 2024. We gratefully acknowledge the individuals who provided information and photographs for this report.

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PREFACE

The Narmada River, often referred to as the lifeline of central and western India, holds unparalleled significance for the region's water resources, ecosystems, and communities. Its hydrological complexity, combined with its ecological and cultural heritage, underscores the critical need for its preservation and sustainable management.

This report, focusing on the identification of threats to biodiversity, management and conservation of the Narmada River Basin, aims to provide an in-depth analysis of the major threats to the conservation of the Narmada River. The collection and compilation of data on the restoration and conservation of biological diversity across the Upper, Middle, and Lower basins will provide valuable insights into the ecological status of the Narmada River Basin.

The findings and recommendations presented in this report are intended to serve as a guiding framework for policymakers, researchers, and practitioners committed to sustainable water management in the Narmada River Basin. By identifying trends and challenges, it aspires to support the development of policies that balance ecological preservation with socioeconomic progress.

We are deeply grateful to all individuals, organizations, and institutions that contributed to the preparation of this report. Their dedication, expertise, and support have been instrumental in shaping this study. It is our hope that this report will inspire collaborative efforts and meaningful action toward the long-term health and sustainability of the Narmada River and its Basin.

Center for Narmada River Basin
Management and Studies (cNarmada)
IIT Gandhinagar, IIT Indore

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List of Abbreviations

WHO	World Health Organization
MoSPI	Ministry of Statistics and Program Implementation
WII	Wildlife Institute of India
UZ	Upper Zone
MZ	Middle Zone
LZ	Lower Zone
NRCD	National River Conservation Directorate
NRCP	National River Conservation Plan
NMCG	National Mission for Clean Ganga
GRBMP	Ganga River Basin Management Plan
CIFRI	Central Inland Fisheries Research Institute
STP	Sewage Treatment Plant
MPPCB	Madhya Pradesh Pollution Control Board
CPCB	Central Pollution Control Board
CWC	Central Water Commission
SBM	Swachh Bharat Mission
NPCA	National Plan for Conservation of Aquatic Ecosystems
CAMPA	Compensatory Afforestation Fund Management and Planning Authority
GPCB	Gujarat Pollution Control Board
NGO	Non-Governmental Organization
NVDA	Narmada Valley Development Authority
ERM	Extension, Renovation and Modernization
mm	Millimetre
km	Kilometre
sq. km	Square Kilometre

1. Introduction

Biodiversity is the variety of life on Earth, including genes, different species, entire ecosystems, natural forests, and man-made landscapes like farms and parks. Biodiversity is an important life-support system that sustains both human health and economic stability. The major ecosystem services provided by biodiversity include soil regeneration, water filtration, and disease regulation (WHO, 2024). That is one of the primary reasons why the global economy is highly dependent on biodiversity. It is reported that around 2.6 billion tonnes of CO2 are absorbed by forests, which help regulate the climate and support other processes, such as pollination, in which pollinators support crops worth over $235 billion. But the actual condition of biodiversity nowadays is deteriorating, and almost 1 million species are currently at risk of extinction. If this rate of biodiversity loss continues, then there will be a huge negative impact on the economy, risk to human health, and livelihoods (WHO, 2024; MoSPI, 2022).

The main objective of this report is to understand the threats to biodiversity and the factors responsible for its loss and degradation, and the management plan for its restoration and conservation.

2. Overview of the Narmada River Basin

The Narmada River Basin extends from 21°40'12" to 23°41'24" N latitudes and 72°48'36" to 81°45'36" E longitudes, encompassing a total area of 98,796 sq. km, which constitutes about 3% of India's total geographical area (3,297,427.32 sq. km). Bounded by the Vindhyan ranges to the north, the Maikala hills to the east, the Satpura ranges to the south, and the Arabian Sea to the west, the basin lies at the northern extremity of the Deccan Plateau and covers significant portions of Madhya Pradesh and Gujarat, with smaller parts extending into Chhattisgarh and Maharashtra. The basin spans 40 districts, comprising 27 in Madhya Pradesh, 7 in Gujarat, 4 in Chhattisgarh, and 2 in Maharashtra. The Narmada River Basin exhibits an elongated shape, stretching about 915.65 km east to west and about 236 km north to south. The basin experiences a tropical monsoon climate characterized by distinct wet and dry seasons, with an average annual rainfall range of 800 to 1200 mm. The Narmada River, the longest west-flowing river in India,

drains into the Arabian Sea near Bharuch, Gujarat, and is further subdivided into the Upper, Middle, and Lower Narmada River Basins as shown in Figure 1.

The Narmada River is characterized by its geographical area, Population, Major Rivers, and Water resources. (Narmada River Basin Atlas 2025). The details are provided in Table 1.

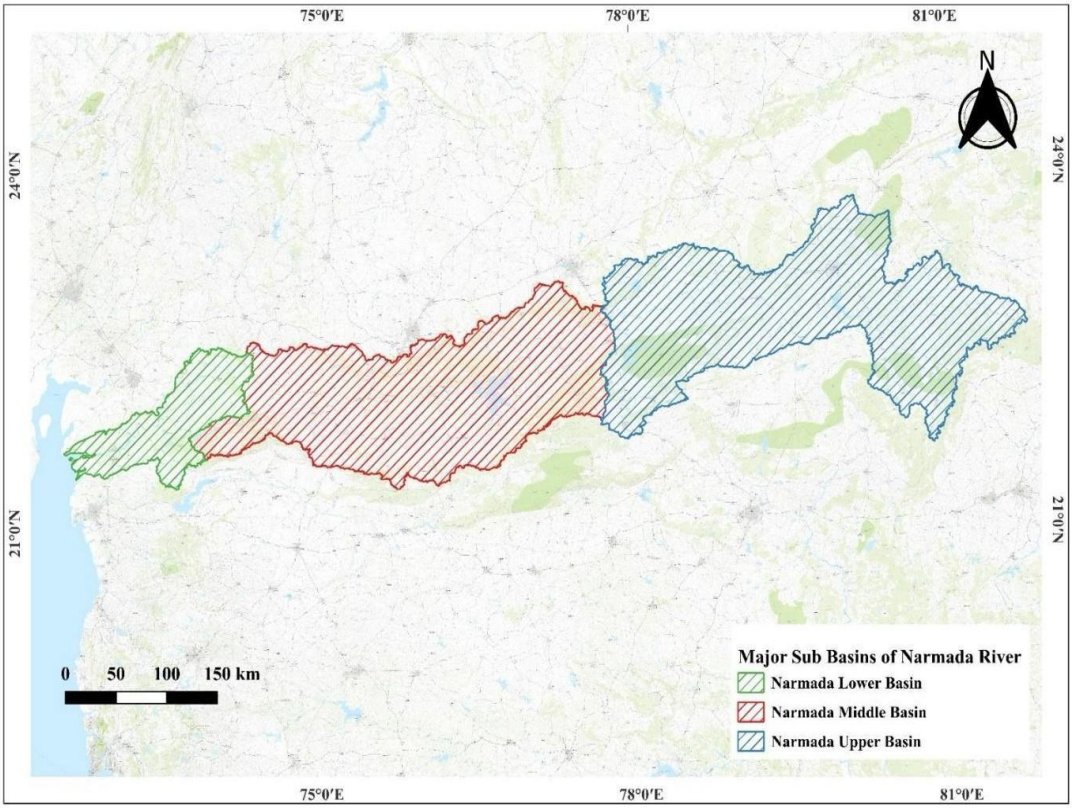


Figure 1: Sub Basins of the Narmada River Basin

Table 1: Salient features of the Narmada River

S. No.	Particulars	Details
1	Basin Extent	Latitude: 21° 40' 12" to 23° 41' 24" N Longitude: 72° 48' 36" to 81° 45' 36" E
2	Total Geographical Area	98,796 sq. km
3	States and their % area contribution in the basin:	Madhya Pradesh (87.82%), Gujarat (9.72%), Chhattisgarh (0.72%), Maharashtra (1.73%)
4	Spatial Variation of Elevation	<5 m: 2.42 %; 5-10 m: 0.15%; 10-50 m: 1.12%. 50-100m: 2.42%; 100-200 m: 11.68%; 200-300 m: 27.26%; 300-400 m: 20.30%; 400-500m: 12.86%; 500-750m: 18.40%; 750-1000m: 3.25%; 1000-1261 m: 0.14%
5	Population	20,799,195 (Census, 2011)
6	Average Temperature Range (°C)	24.92 - 27.39
7	Mean Maximum Temperature (°C) (1970-2023)	32.47
8	Mean Minimum Temperature (°C) (1970-2023)	19.01
9	Mean Annual Rainfall (mm) (1970-2023)	1110.83
10	Highest Elevation (m)	1261
11	Major Rivers	Hiran, Tendoni, Barna, Kolar, Man, Uri, Harni, Orsang, Burhner, Banjar, Sher, Shakkar, Dudhi, Tawa, Ganjal, Chhota Tawa
12	Districts	39
13	Number of Water Resources Structures	Dams-277; Barrages-2; Weirs-2; Lifts-4; Power Houses-9

14	Highest Dam	Sardar Sarovar Dam-163 m
15	Longest Dam	Rani Avanti Bai Sagar -Bargi (NVDA) Dam-5.357 km

16	Highest Barrage	Kolar Barrage-15.5 m
17	Longest Barrage	Kolar Barrage-1.195 km
18	Number of Irrigation Projects	Major-21; Medium-23; ERM-1
19	Number of Hydroelectric projects	6
20	Number of Groundwater Observation Wells	393
21	Number of Hydro-Observation sites	29
22	Number of Flood Forecasting Sites	4
23	Water Tourism and Sport Sites	27

Source: Central Water Commission and National Remote Sensing Center, Ministry of Jal Shakti, Government of India (2014a). Basin Report of the Narmada River Basin.

3. Tributaries of the Narmada River Basin

The important tributaries of the Narmada River are the Burhner, the Banjar, the Sher, the Shakkar, the Dudhi, the Tawa, the Ganjal, the Kundi, the Goi, and the Karjan, which join from left, whereas the Hiran, the Tendoni, the Barna, the Kolar, the Man, the Uri, the Hatni, and the Orsang join from right. Narmada drains into the Arabian Sea through the Gulf of Khambhat. Details of the tributaries are provided in the flowchart in Figure 2.

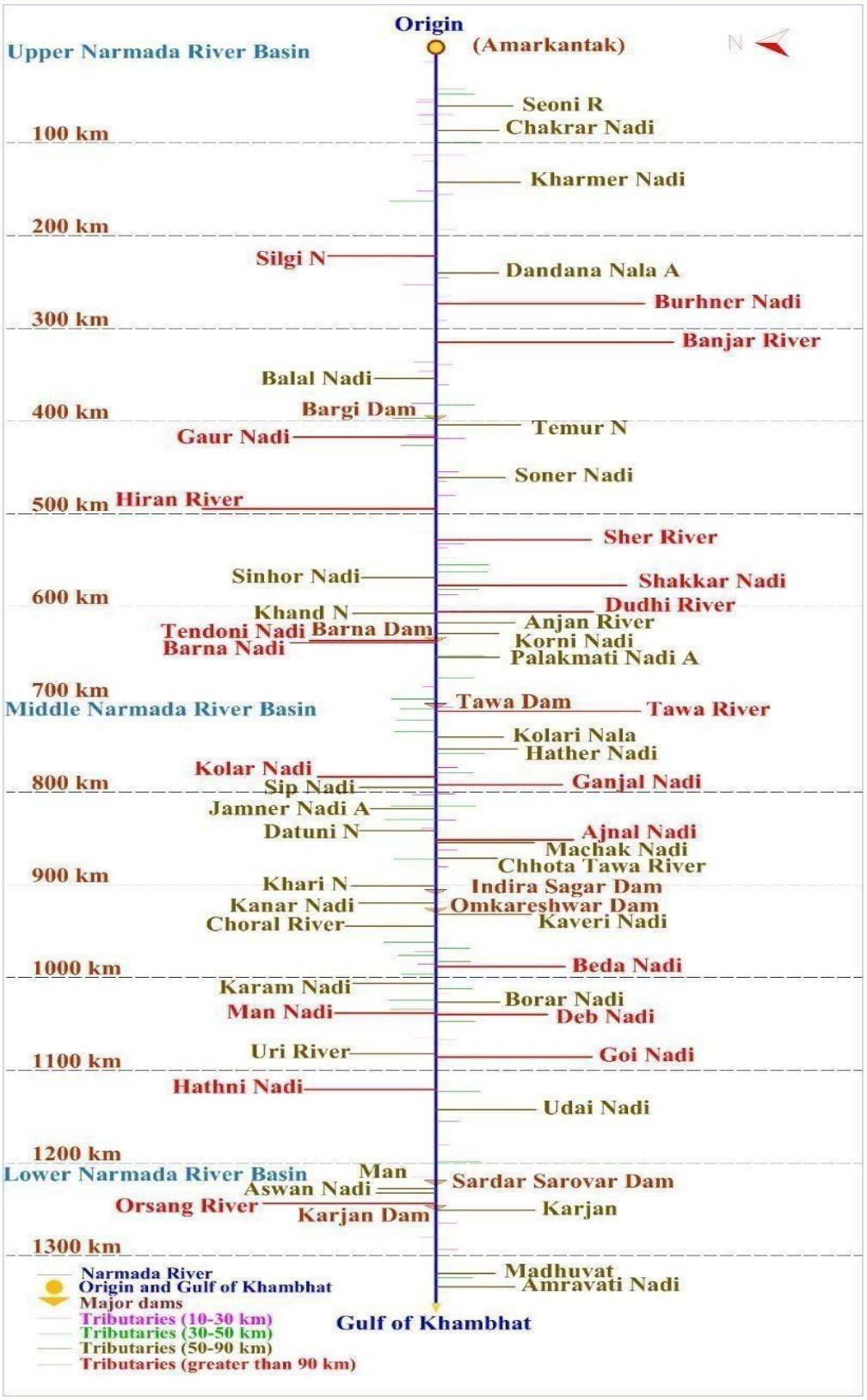


Figure 2: Flow Diagram: Narmada River and its Tributaries

Source: River Atlas, 2025 (cNarmada) IIT Indore.

4. River Zoning

As per the (WII 2022) report, the Narmada River is classified into three major zones:

a) Upper Zone (UZ): Amarkantak and Bheraghat

b) Middle Zone (MZ): Bheraghat and Chandod

c) Lower Zone (LZ): Chandod in the Vadodara district to the Gulf of Khambhat

As part of biodiversity studies and management plans for restoration and conservation, information on tributaries and river zoning is very important for research studies.

5. Biological diversity of the Narmada River

Biological Diversity of the Narmada River Basin includes a wide variety of flora and fauna. The Narmada River and its tributaries support different types of aquatic life as part of the ecosystem. Variety of flora in the Narmada River Basin includes different classes of phytoplankton, periphyton, macrophytes, riparian, and floodplain vegetation, and, in the case of fauna, the Narmada River consists of a wide variety of aquatic species, which include zooplankton, macrobenthos, fish, higher aquatic vertebrates, and aquatic aves.

The biodiversity of the Narmada River Basin is remarkably rich, with a vibrant array of flora and fauna. Flora includes various phytoplankton and periphyton species in the region, which contribute to a diverse plant life across all sub-regions of the Narmada River Basin. There are many common species of flora and fauna in the sub-basins, shows high ecological connectivity within the river ecosystem.

Among fauna variety of fish species are common in the upper, middle, and lower basins, demonstrating strong ecological interrelationships, some of the species that are particular and unique to specific sub-basins are also present and shows specific habitat requirement. The lower tributaries of the Narmada River support a rich and dynamic ecosystem, but they are also highly vulnerable to environmental pressures. The macrobenthos community act as a indicator of environmental system, showing a gradual transition from diverse freshwater organisms in the upper regions to pollution or stress tolerant annelid species communities in the industrialized estuarine area. The river's fish diversity id particularly remarkable , with the record of 84 taxa highlighting the importance of the Narmada River and basin importance

for vital migration route and habitat for numerous fish species. Overall, the Narmada River Basin serves as a critical ecosystem, supporting a wide range of life forms and highlighting the interconnectedness of its environmental components (Biological Profile Report on Narmada River Basin, c Narmada, 2025).

6. Threat Identification

According to the WHO data there is rapid decline in the biodiversity. The species extinctions rate is 10-100 times higher than the natural baseline. Major cause of these alarming results is due to human actions such as deforestation,industrialization, habitat fragmentation, and climate change.

Biodiversity in the Narmada River Basin is increasingly threatened by anthropogenic pressure, which includes rapid developmental activities, unsustainable resources, and industrialization. Major threats to biodiversity in the Narmada River Basin include land-use and land-use change, which leads to habitat loss and overexploitation of aquatic resources, including fishing and poaching. Additional challenges arise from sand mining, aquatic pollution due to sewage discharge, industrial wastewater, and developmental projects in cities, climate change, and the introduction of invasive species, as shown in Figure 3.

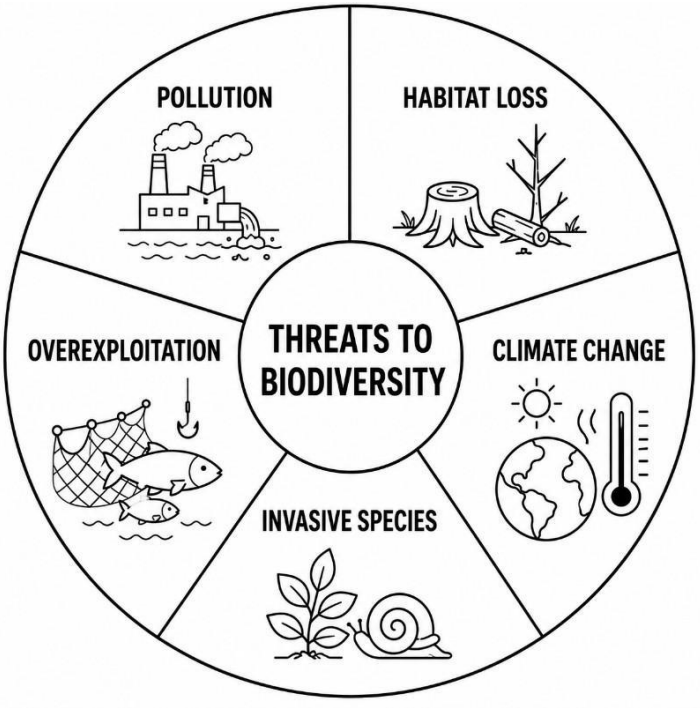


Figure 3: Types of threats to Biodiversity

6.1. Habitat Loss and Fragmentation

Habitat loss is defined as a decrease in the number of habitats available to an organism, resulting in a reduced capacity of the environment to sustain the wildlife population (Fahrig et al., 2003).

Habitat fragmentation is the process by which large, continuous habitats are broken into small, isolated patches, leading to habitat loss. This fragmentation and habitat loss are mainly due to human activities such as deforestation, urbanization, and agriculture. While some habitats are naturally patchy, anthropogenic activities have increased fragmentation, affecting habitat quality (Haddad et al., 2015).

Habitat transformation is the leading cause of biodiversity loss, especially in tropical regions, which have more than 75% of all species. Habitat loss and fragmentation are primary threats to biodiversity globally, but their consequences are typically complex (Gann et al., 2021).

In the Narmada River Basin, there are a total of 277 dams, including the big projects Indira Sagar and Omkareshwar dams in the Middle Narmada River Basin, and the Sardar Sarovar Dam in the Lower Narmada River Basin. In one study on the Indira Sagar and Omkareshwar dams, it was reported that both submerged 45,609 hectares of forest, resulting in the fragmentation of pristine forests. This fragmentation leads to reduced habitat, degraded habitat quality, and increased extinction risk for many species (Tata, 2021).

Data on land-use and land-cover changes in the upper catchment area of the Narmada River Basin have become a major cause of ecological degradation and habitat loss. Past 20 years, there are significant changes in land use and land cover have occurred in this region, due to urbanization, infrastructure development, population growth, and climate change. The data analysis shows that there is a decline in forest land and fallow land, which is around 3.21% of the total area. There is a 300% increase in built-up areas, driven by the conversion of agricultural land, forests, and water bodies. Because of these changes, habitat loss has increased and negatively impacting biodiversity and the ecosystem. Also, uneven terrain, different soil conditions, and various socio-economic activities in the upper catchment region of the Narmada River Basin make this region more vulnerable to ecological disturbance (Patel et al., 2024).

6.2. Aquatic Ecosystem Pollution

It is important to understand the aquatic pollution in the Narmada River, as it ultimately affects the entire biodiversity and human health. There are various kinds of

pollutants present that directly or indirectly affect the entire river ecosystem. These include emerging contaminants such as heavy metals and microplastics/nanoplastics in river water, other pollution load factors are sewage water or domestic discharge, industrial discharge, agricultural practices include the use of fertilizers and pesticides, sediment influx, and other developmental, religious, and tourist activities.

Across the Narmada River Basin, various heavy metals, including iron, chromium, zinc, and nickel, are present in water at varying concentrations. One example is Kapil Dhara at Amarkantak, the origin point of the Narmada River, which exhibits high levels of iron and aluminum, with surface runoff and rock weathering as the major causes. Another location is Dindori, which shows a high level of manganese and chromium, likely from agricultural runoff and untreated domestic sewage. In the upper basin itself, Jabalpur also showed high lead concentrations, which may be due to diesel boats or urban activities (Gupta et al., 2025). Another significant contaminant in the Narmada River is microplastics less than 5mm in size. Recent studies by Patidar et al. (2024) at Maheshwar and Omkareshwar, which are part of the Middle Narmada River Basin, reported microplastic pollution in water and sediment samples from areas with high residential and tourism activity. Polystyrene and polyvinyl chloride were predominantly found in the water samples and sediments. As is well known, both locations have high levels of ritual waste, fishing, and tourism, which may be one of the reasons for microplastic contamination in water samples and sediments. The primary sources for microplastics include household sewage, single-use plastics, tissue waste at ghats, and fishing equipment. Many studies have shown the disruptive impact of microplastics on aquatic biodiversity, and, in addition, these microplastics serve as carriers for heavy metals, intensifying ecological risks.

The next category responsible for aquatic pollution is the condition of sewage treatment in the basin's cities. The Narmada River faces a high pollution load from untreated sewage and industrial discharges. Many cities, like Jabalpur and Khandwa, generate more sewage than they treat, and cities like Narmadapuram still don’t have operational STP, leading to direct discharge into the river (MPPCB 2025). In industrial areas, cities like Betul and Jabalpur, a major urban Center, have inadequate common effluent treatment plants. Industrial effluent, notably from the industrial area near Bharuch and Ankleshwar of the lower Narmada River Basin, introduces toxic chemicals into the river (Hussain et al., 2014). These untreated discharges into the river result in high levels of biological and chemical contamination, posing a danger to both the ecosystem and public health. Cities like Narmadapuram, Balaghat, Alirajpur, and Umaria lack

operational STPs, while others, such as Jabalpur and Khandwa, have inadequate capacity, leading to the direct discharge of untreated sewage into the river as mentioned in Figure 4. Data sourced from the District Environmental Plan (MPPCB, 2025, SBM 2.0, 2026, and MPPCB water quality reports (2024-2025)).

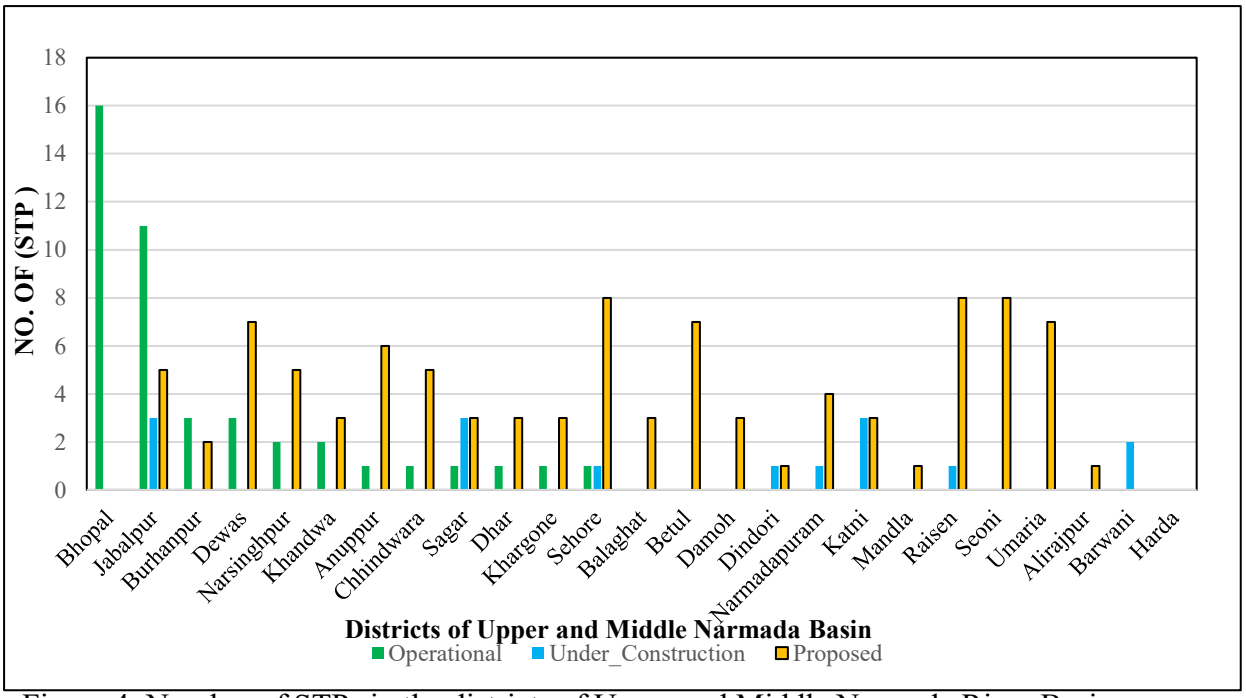


Figure 4: Number of STPs in the districts of Upper and Middle Narmada River Basin

The next category of pollution contributors is agricultural runoff and sediment. Major districts such as Khandwa, Sehore, Barwani, Dindori, Mandla, and Jabalpur are agriculture-based districts and are major contributors to farming in the basin. Due to extensive farming practices and overuse of fertilizers and pesticides, the river water quality and ecosystem are affected. As the major crops of these regions are soybean, wheat, and sugarcane, these crops require a high amount of fertilizers, which leads to increased levels of nitrogen and phosphorus. These increases in nutrients contribute to the agricultural runoff, which enters the Narmada River directly and to its tributaries, such as the Tawa and the Sher. Farming practices along the Riverbank, at the time of monsoon, lead to large quantities of nutrients and sediments entering the river(CPCB Water Quality Reports, 2022; Narmada River Basin Agriculture Profile Report, 2025, Central Pollution Control Board, 2019).

Another factor affecting the biodiversity of the river is the sediment load, and the contributors are farming on sloped land, deforestation, sand mining, and land disturbance in the upper zone of the Narmada River. Sediment transport in the river system primarily occurs in the monsoon season and affects the habitat in the river ecosystem. As the increasing amount of sediment will reduce the penetration of the sunlight, which affects the growth of aquatic plants and habitats, and will disturb the life cycle. There are also sediment data with the detection of heavy metals such as manganese, cadmium, lead, and nickel for cities like Dindori, Mandla, and Jabalpur. These heavy metals are toxic and can accumulate in the tissues of aquatic organisms, and through the process of biomagnification, can also enter the food chain. This accumulation is not only dangerous to fish and other aquatic organisms but also to the human population, increasing the risk of health disorders. (Rahi et al., 2024; Nutrient and Sediment Load Report, 2025).

There is constant increase in the sediment load, riverbank erosion, alteration in river geomorphology due to sand mining specifically in Narmadapuram. All these factors will affect **biodiversity and water availability (Singh et al., 2026).**

The accumulation of nutrients and sediments in the Narmada River leads to algal blooms, eutrophication, and degraded water quality. Locations like Kabir Sarovar and Kharighat, where macrophyte growth is excessive, exhibit high organic and nutrient loads, leading to decreased dissolved oxygen and ultimately affecting aquatic biodiversity (Saini et al., 2025).

Addressing these issues requires promoting sustainable agricultural practices, precision irrigation, and effective soil and water conservation measures to reduce runoff and sedimentation, thereby protecting the Narmada River's ecological health.

Religious activities at various ghats on the Narmada River, such as Riddhnath Ghat (Harda District), Mandla, and Narmadapuram (Sethani and Budhini Ghats), significantly impact water quality due to mass bathing and ritual offerings. Due to these activities, there is a massive increase in organic and inorganic pollution, the introduction of pathogens, and alterations in key water parameters such as dissolved oxygen and turbidity (Rathore et al., 2023; Gupta et al., 2020). The arrival of pilgrims and tourists at sites like Bhedaghat and Omkareshwar further exacerbates pollution.

In brief, the aquatic ecosystem of the Narmada River is being disturbed by various threats, including heavy metals, microplastics, sewage and industrial discharges, nutrient and sediment

loads, religious activities, etc. The major zones facing these issues are Jabalpur, Mandla, Dindori, Omkareshwar, Maheshwar, Narmadapuram, and Bharuch. To solve these issues, there should be proper management of agricultural practices, solid and liquid waste management, and regulation of religious and tourism activities, so that River health is restored and maintained for the aquatic ecosystem.

6.3. Invasive Species

According to the data available, there are 106 invasive alien species of plants that have been recorded. In the Narmada River Basin, a total of 5 Ichthyfauna are recorded, of which one is present in the upper zone, and the remaining 4 are present in the middle Zone. Data recorded for fish in the Narmada River reveals that small indigenous fish are more abundant than the fish that are economically important. The small fishes that are indigenous are *Osteobrama cotio, Chada nama, Systomus* sarana, *etc*. The fish species that are economically important in the Narmada River are *Labeo Calbasu, Wallago attu*, etc. This increase in indigenous species is due to habitat degradation, overfishing, and the introduction of invasive species. As it is known that small fishes are not that much contributors to the fisheries and will also have an impact on the focal species or species which are important in terms of conservation, like *Wallago attu*, which is in the vulnerable category (WII, NRCD 2026). A few names of the invasive species are provided in the Table.

Table 2: Invasive species in the Narmada River Basin (WII NRCD,2026 Report)

Plants Species	**Fish Species**
Lantana Camara, Parthenium hysterophorus, Eichhornia crassipes, Prospis juliflora	*Cyprinus carpio, Hypophthalmichthys nobilis, Hypophthalmichthys molitrix, Oreochromis mossambicus and Oreochromis niloticus.*

7. Effects on biodiversity

The Narmada River has an abundant biodiversity of fish. A total of 167 and 157 species have been found in the Upper and Middle basins of the Narmada River, of which 114 fish species are common to both basins (Biological Profile Report of Narmada, 2025). Over time, the fish fauna has been exploited. In 1958 and 1966, 269 tons of catch were recorded from the Narmada River, which increased to 11.1% of total production in 1972. As per the Narmada River

Development Authority, about 3474 tons of fish have been harvested from various reservoirs situated within the Narmada River Basin. (WII 2022).

Based on the study by CIFRI, it was concluded that the construction of the Sardar Sarovar Dam in Gujarat affected the natural flow of the Narmada River, resulting in reduced freshwater discharge and irregular flow patterns. Due to these changes, the Hilsa (*Tenualosa Ilisha*), which is a migrating fish of the lower estuarine region of the Narmada River, was adversely affected. Data before dam construction paints a different picture than data on hilsa production after dam construction. In 1974-75, Hilsa contributed about 977.1 tons. However, after dam construction, a continuous decline was observed, from 3448 tons in 2004-5 to 419 tons in 2014-2015, representing a drastic decrease in an economically important fish.

Another important change observed is a shift in fish species in the estuary. Before the dam, carp, catfish, and prawns were dominating, but afterward, mullets and mudskippers became more prevalent.

In the lower stretch of the Narmada River, a decline in the hilsa fish was observed, for which one of the major reasons is overfishing, especially the catching of juvenile hilsa by using small, meshed bag nets. (Bhakta D et al., 2018)

Based on secondary data analysis, there has been a major decline in fish, particularly Mahaseer. The main reasons are the use of illegal and harmful fishing methods, such as dynamite and agricultural pesticides. These practices have been observed mainly in the region from Gadarwada to Harsud and up to Surpan, in addition to the use of a disco net, a very fine-mesh net used for catching small, immature fishes like mahaseer and major carps. All these factors equally contributed to the decline of the fish fauna in the Narmada River. (Central Inland Fisheries Research Institute 2009).

8. Restoration and conservation Plan for the Narmada River Basin

To restore biodiversity, land restoration, and climate change, there are several worldwide conventions and frameworks that focus on all the important aspects of ecosystem management. These include the Paris Agreement, which aims to mitigate climate change by focusing on forest and agricultural conservation and climate

adaptation measures. Another global convention is the Kunming-Montreal Global Biodiversity Framework, which aims that till 2030 less biodiversity loss, restore degraded ecosystems, and achieve land and water conservation by at least 30%.

After independence, India also focused on the importance and protection of the environment. The constitution of India emphasizes and highlights how important the environment, forests, and wildlife are for a healthy planet. To support these, there are several policies introduced by the government for biodiversity conservation and restoration. Some of the important initiatives are the Biological Diversity Act, which promotes the sustainable use of biological resources and biodiversity conservation; the Forest Conservation Act (1980) and the Wildlife Protection Act (1972), made for protecting forests and wildlife. Another effort is the Green India Mission, which comes under the National Action Plan on Climate Change (2008), aiming for afforestation and ecosystem restoration. There are two more policies that specifically focus on ecological restoration and sustainable natural resource management across the country: the Compensatory Afforestation Act (CAMPA 2016) and the National Action Plan to Combat Desertification (NAPCD 2023).

There are several other implementations done by the government of India for the restoration and conservation of biodiversity. These projects are Project Tiger, Project Elephant, Amrit Dharovar Initiative, and many more. These projects and initiatives reflect India’s strong commitment to environmental protection in terms of ecosystem restoration and biodiversity conservation.

Among many schemes, there is also a scheme by the Ministry of Environment, Forest, and Climate Change under the Wetlands Division. This scheme, named the National Plan for Conservation of Aquatic Ecosystems (NPCA), focuses on holistic conservation of wetlands so that there is improvement in the water quality, biodiversity, and ecosystems get benefits. This scheme's main aim is to include this wetland in developmental programs by supporting management plans, training programs, and research work (Rahmi & Anand, 2024).

For the conservation of biodiversity in the Narmada River Basin, conservation tools are categorized into scientific, legal, and traditional tools. The aims of these approaches are to restore and conserve the water quality, biodiversity, traditional heritage, or entire ecosystem in support of stakeholders, participation, government, and scientific solutions.

Scientific tools focused on research on biodiversity, keeping control of the pollution levels, genetic composition of species, etc. The pollution level in the Narmada River is monitored by the two state government bodies, i.e., Madhya Pradesh Pollution Control Board (MPPCB), Gujarat Pollution Control Board (GPCB). Continuous monitoring of the biodiversity and pollution levels is important for effective conservation. The next approach for conservation is legal tools. In Madhya Pradesh and Gujarat, there are several policies and acts made for the restoration and conservation of the wildlife, fisheries, water, etc., which play an important role in the maintenance of a healthy planet. Details of the State policies and Acts are given in Table 2.

Legislative tool at the state level for conservation of the River Narmada and its Biodiversity (WII 2022).

Table 3: State-level Legislation and Policies

State	Legislations/Policy	Key Features
Madhya Pradesh	State Water Policy, 2003	Focuses on sustainable water resource management. Prioritizes water for drinking, irrigation, afforestation, and power generation.
Madhya Pradesh	Fisheries Act, 1948	Focused on the development and protection of the fish's life by prohibiting the use of explosives and chemicals in the river, controls and regulates the construction of dams and weirs. Prohibition of throwing solid and liquid waste into the river, which causes the death of the fish.
Madhya Pradesh	Sand mining rules, 2018	Regulates the sustainable use of sand and controls sand mining near the cultural and heritage sites of importance. Requires environmental clearance for the sand mining activities.
Madhya Pradesh	Mineral policy, 2010	Ensure the sustainable use of mineral resources, especially those in the river valleys like the Narmada
Gujarat	Water policy (2015) draft	Emphasizes integrating water resource management, ecological needs, climate resilience, prioritization of water use for drinking, irrigation, fisheries, hydro power, and industries.
Gujarat	Irrigation and Drainage Act, 2013	It defines penalties for polluting canals or Rivers with any solid or liquid pollutant and damaging

		embankments or constructions to change the river flow.
Gujarat	Water User's Participatory Irrigation Management Act, 2007	Provides a water users association for water resource management, and also penalizes unauthorized dams or any abstractions that affect the river flow and embankments.

The third approach is the traditional tool in which several rituals, beliefs, holy places, and mythological puranas are part of the river conservation. Traditional conservation of the Narmada River depends on the people's cultural practices and religious beliefs. For the population, the Narmada River is not just a river; it is worshipped as a sacred river, as a mother. These religious believe makes strong connection between the river and society to keep it healthy and clean. People who perform the Narmada Parikrama, a pilgrimage around the river, perform rituals like holy baths, water offerings, and celebrations on the ghat of Narmada Jayanti. People worship stones from the river as shivlings and perform many religious activities at the riverbanks. Places like Amarkantak, Maheshwar, and Omkareshwar have different levels of importance as holy places along the Narmada river bank. All these rituals and religious activities help in making a strong connection between the people and the river, which helps in conservation and protection of the river, which ultimately affects the biodiversity and ecosystem.

There are many sacred trees and forests along the riverbank, which are protected and worshiped by the community. These trees and forests provide habitats to a wide variety of species, thereby contributing to biodiversity conservation. Another important aspect is eco tourism and archeological sites like Dhurandhar falls and Kapil Dhara falls, which promote the ecological and cultural importance of the river among the people. All these rituals, cultural activities, and tourism play an important role in protecting the Narmada River as they connect the people with the river and allow them to understand the importance of the river's health and the life associated with it. Ultimately, the results of this connection and awareness among the communities are for the biodiversity and the environment.

Despite many policies, acts, and conservation tools, the water quality, e flow, and biodiversity of the Narmada River and Basin are in danger; scientific studies on the basin's flora and fauna are very limited and restricted to specific areas only. Studies have

been done so far on regions such as the Achanakmar-Amarkantak biosphere, Shoolpaneshwar Wildlife Sanctuary, which lies in the upper basin (Singh et al., 2010; Sharma et al., 2010). There is very less information, and data is available on the riparian and aquatic vegetation of the Narmada River Basin. Research is needed on the pollution level, as well as the impacts of the different pollutants and anthropogenic activities on the ecosystems.

Similarly, studies on fish diversity of the Narmada River Basin is also very limited. Due to an increase in anthropogenic activities and pollutants, there is a requirement for continuous monitoring of the fish diversity of all three sub-basins, as there are very few data available for the upper, middle, and lower basins. More research and study are required for flora and fauna for the restoration and conservation of the biodiversity of the Narmada River Basin. Research has been done on heavy metal contamination, but there are more emerging contaminants like pharmaceutical compounds and microplastics, which are affecting the aquatic ecosystems and human health, and need more research and suitable action for their removal. The focus on threatened species also requires, due to increasing human pressure, genetic studies needed for understanding the dynamics of their population and survival capacity in this situation. In short, there is strict attention needed in terms of studies and research to understand the dynamics of the population of flora and fauna in various regions of the Narmada River Basin, and also studies and research should focus on the various pollutants that have a harmful impact on the biodiversity and ecological health.

Aside from this knowledge gap, there is another issue that, despite many acts and policies, biodiversity and ecosystems are not in good condition. The main problem is the lack of coordination between the government policies and local communities. Opposition to the Sardar Sarovar dam, which is a large-scale project, is evidence of this lack of coordination and support by the local communities. For the success of any government policies or acts, support and collaboration are required between the government, NGOs, and the local people. For the sustainable restoration and the conservation of the biodiversity of the Narmada River Basin, a framework of active governance, community involvement, and science-based solutions is required (WII 2022).

9. Recommendation for Conservation of the Narmada River.

1. The establishment of a confederation of Regional Institutions is proposed to maintain, restore, and conserve the biodiversity of the Narmada River Basin. The consortium includes the following government bodies: forest departments, pollution control boards, water resource departments, research institutes, NGOs, and local communities in Madhya Pradesh and Gujarat. The role of these bodies will be to coordinate conservation actions, data sharing, and policy implementation to sustain the river's health.
2. Maintaining the natural environmental flow is the most important factor for the protection of flow patterns, riparian ecosystem, aquatic life, and biodiversity of the river. It is the responsibility of the Central Water Commission to ensure the minimum flow of the Narmada River, and for that, water audits should be done in support of the Central Water Commission and the water resource departments of the states.

3. Treatment of domestic and industrial discharge before mixing into rivers or their tributaries is very crucial for the river ecosystem. The data show that sewage waste is generated, but there is no proper infrastructure (STP) for its treatment in the Narmada River Basin. Municipalities should focus on preventing unauthorized waste dumping and increasing the number of STPs, especially in cities along the Narmada River. So, the very first requirement is to develop infrastructure to treat domestic and industrial effluents for a sustainable river. There should be an upgrade in the treatment plant technology for industrial and domestic discharges that effectively remove heavy metals, microplastics, and other pharmaceutical chemicals from water. The pollution control board can measure water quality at regular intervals and ensure compliance with rules and regulations.

4. There should also be more promotion and outreach activities done regarding the agricultural practices to reduce the use of pesticides and fertilizers, and these chemicals affect the aquatic life of the river, which ultimately affects the entire ecosystem. Policy measures should strictly focus on controlling the use of pesticides and fertilizers.

5. To restore habitat in the Narmada River, it is important to reconnect the fragmented river stretches so that there is proper maintenance of the flow for the migrating species. There should be reforestation along the riverbank with native species. After the planting of riparian vegetation, proper monitoring should also be conducted to control invasive plant species. The forest departments of Madhya Pradesh and Gujarat should take this step of planting riparian flora along the Narmada riverbank to increase green cover. Apart from flora, fauna should also be regularly monitored by the forests and wildlife departments so that the effects of anthropogenic activities on biodiversity can be assessed and, based on that, a conservation plan can be developed.

6. There is a strict requirement for the regulation of anthropogenic pressure, such as fishing, sand mining, ferry boats, etc. For sustainable fishing, regulations such as gear restrictions, catch limits, and seasonal closures should be implemented and enforced. In the Middle and lower zones of the Narmada River, regular monitoring of sand mining should be done to prevent habitat destruction. Tourist activities, the use of ferry boats, their speed, and engine noise should all be regularised to preserve biodiversity.

7. Community Awareness and Engagement should also be a part of restoration and conservation measures. Conduction of regular workshops, educational school activities, and stakeholder workshops, local people awareness programs regarding the implementation of conservation planning. Information boards can be placed at tourist sites and ghats along the banks of the Narmada River to sensitize local people about the importance of a clean river, its biodiversity, and life.

8. Policy and Legal Framework are the dynamic parts of any plan. In restoration and conservation planning, existing plans should be properly implemented, and policies should be regularly reviewed and updated based on monitored data and emerging threats.

10. Conclusions

The ecological and biodiversity assessment of the Narmada River uncovers that it is the most significant river in terms of ecology and culture. The Narmada River is home to 469 plant species, more than 145 species of amphibians, reptiles, birds, and mammals. The Narmada River Basin includes a variety of habitats, ranging from forest to wetlands to agricultural land. The Narmada River supports a large population through water supply, agriculture, fishing, and many other benefits.

However, due to rapid urbanization, industrialization, agricultural expansion, dam construction, sand mining, and the introduction of invasive species, the Narmada River faces considerable threats to its biodiversity. Despite the existence of many policies, acts, and protected areas, the issue of biodiversity loss remains. There is very limited engagement of the public, highlighting the need for collaborative, science-based conservation strategies that involve all stakeholders. There is a strong need to restore and conserve the biodiversity in the Narmada Basin.

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